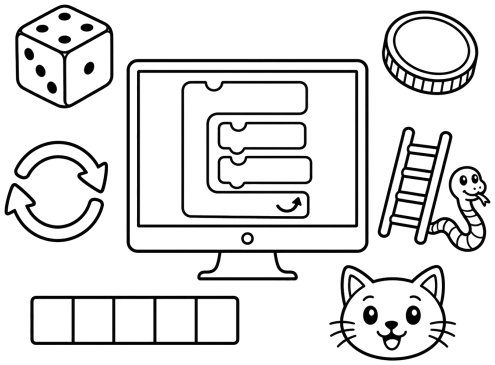
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 9



UNIT 3
Programming

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Choose between two options with simple selection.
- [ ] Choose one option out of many with IF THEN ELSE blocks.
- [ ] Draw and read flowcharts that repeat steps.
- [ ] Use the repeat, repeat until and forever blocks in Scratch.
- [ ] Build programs with a repetition inside a repetition.
- [ ] Store, add and show many items with an array (a list).
- [ ] Divide a problem into sections and build its program.

THREE RULES FOR THE COMPUTER ROOM

- **Always** save your Scratch project with a clear name, and save it often.
- **Never** download or install software without asking your teacher.
- **Never** eat or drink near the computer.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the practical task on a computer with Scratch. Last you **write** the exercises in this book.

Colour the small pictures when you finish your work early.



Colour me!

THE WORDS YOU WILL LEARN

Selection	Simple selection	Condition	Decision symbol
IF THEN ELSE	Repetition	Control structure	repeat
repeat until	forever	Nested repetition	Variable
Multiple	Random number	Pen	Array
List	Data structure	Item	length of
add	Make a List	Average	Problem analysis

"A program that can choose, repeat and remember can do almost anything."

- THE KING OF ICT

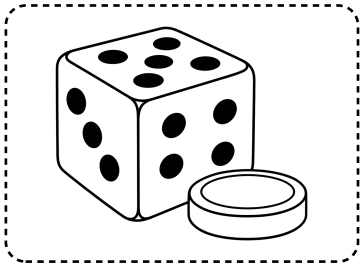
CHAPTER 1

Selection

? MAKE YOUR OWN GAMES

"I love computer games!" "Games are not bad. But it is better to develop your own games."

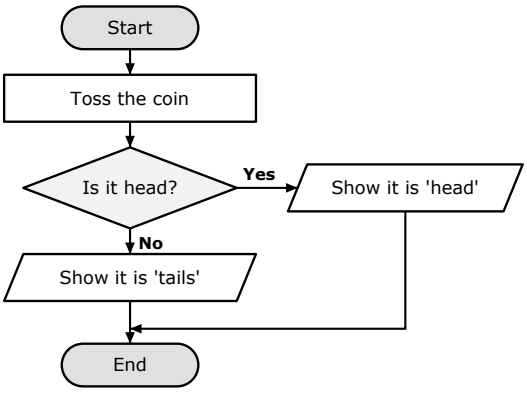
Every game must **choose**: head or tail? Which number on the dice? In this chapter your Scratch programs learn to choose.



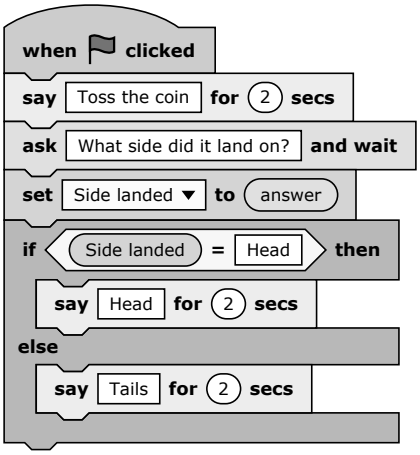
Colour me!

3.1 Simple selection

In Grade 8 you learnt that a **simple selection** selects **one option out of two** options under a certain **condition**: for example "head" or "tail" when we toss a coin. Simple selection has **one condition** with **two options**. If the condition is **true**, one option is selected. If not, the other option is selected.



Flowchart 1: "head" or "tail" of a coin



Scratch program 1 (Side landed keeps the answer)

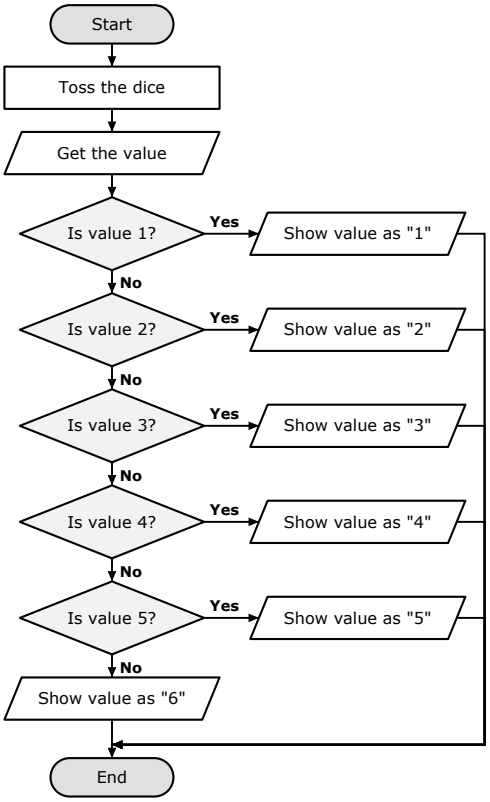
> DO IT - PRACTICAL 1 (WORKBOOK ACTIVITY 3.1)

Make the head or tail program (Scratch program 1). **Ask your teacher first.**

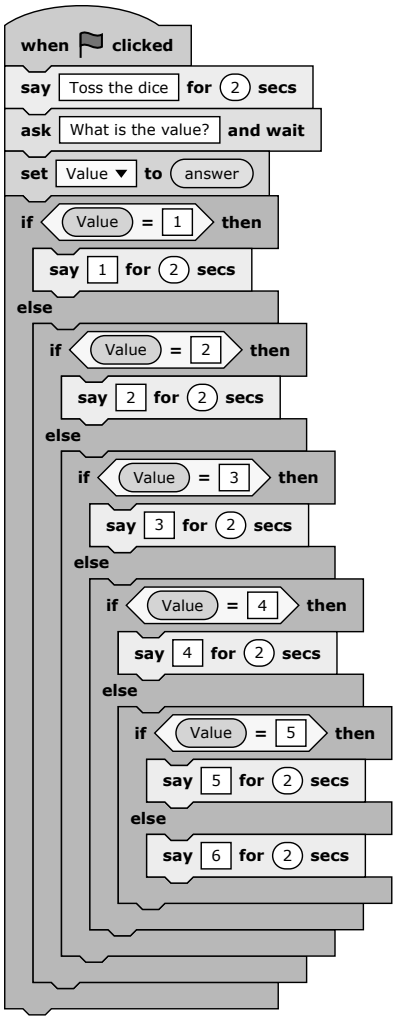
1	Open Scratch. Click **Data** and make a variable called **Side landed**.
2	Drag **when green flag clicked** (Events) and **say Toss the coin for 2 secs** (Looks).
3	Add **ask What side did it land on? and wait** (Sensing), then **set Side landed to answer**.
4	Add **if ... then ... else** (Control) with **Side landed = Head** (Operators). Put **say Head** in the first part and **say Tails** after **else**.
5	Toss a real coin. Click the green flag and type what you got: **Head** or **Tail**. Save it.

3.2 Selection out of many options

Selection out of many options selects **one option out of more than two** options. Example: tossing a dice for the values 1 to 6.



Flowchart 2: the value of a tossed dice



Scratch program 2

The conditions are checked **one by one**.

A value is shown when **one** of the five conditions is satisfied.

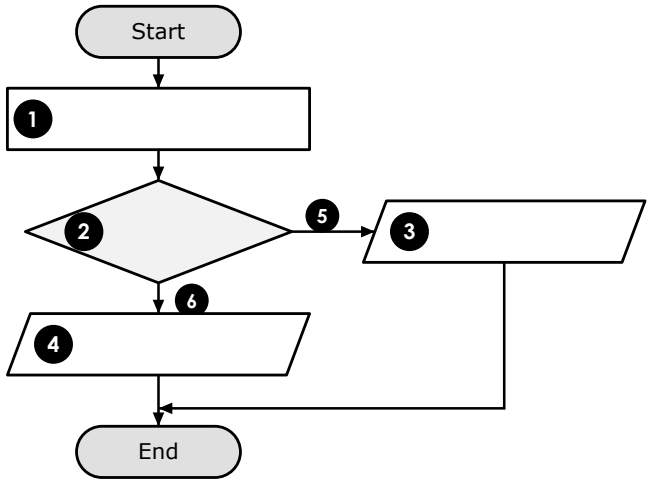
When **none** of the five conditions is satisfied, the value 6 is shown.

In Scratch, each **if ... then ... else** block goes inside the **else** part of the one before it.

A Complete the flowchart

6 marks

Flowchart 1: write the words in the numbered symbols and on the arrows 5 and 6.



USE THESE

- Show it is 'head'
- Yes
- Toss the coin
- No
- Show it is 'tails'
- Is it head?

B Fill in the blanks

6 marks

Use the words in the box. Two words are extra.

WORD BANK

two	one	more than two	true
else	decision	repeat	six

1. Simple selection selects one option out of ..... options.
2. Simple selection has ..... condition.
3. Selection out of many options selects one option out of ..... options.
4. If the condition is ..... , the first option is selected.
5. In Scratch, the other option goes in the ..... part of the IF THEN ELSE block.
6. In a flowchart, the condition is written in the ..... symbol.

C What will the cat say?

6 marks

Look at Scratch programs 1 and 2. Write what the cat says.

Scratch program	You type	The cat says
1 (the coin)	Head	
1 (the coin)	Tail	
2 (the dice)	2	
2 (the dice)	5	
2 (the dice)	6	
2 (the dice)	9	

D Answer these

4 marks

Write full sentences.

1. What is the difference between simple selection and selection out of many options? (2 marks)

2. Scratch program 2 does not check "Value = 6". Why is it not needed? (2 marks)

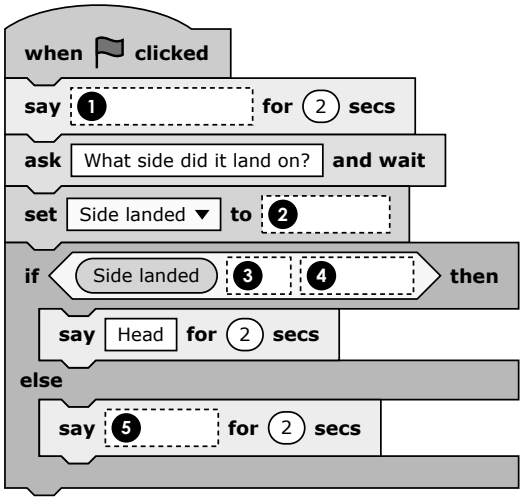
E Complete the script

5 marks

Scratch program 1: write the words for the numbered gaps. Two words are extra.

WORD BANK

answer	Tails	=	Toss the coin
Head	Value	>	



1. .....
2. .....
3. .....
4. .....
5. .....

F Simple or many options?

6 marks

Tick the kind of selection.

Selection	Simple selection	Many options
Head or tail when you toss a coin	☐	☐
The value of a tossed dice: 1, 2, 3, 4, 5 or 6	☐	☐
One condition with two options	☐	☐
Selecting one option out of more than two options	☐	☐
Is the student present? Mark 1 or mark 0.	☐	☐
Five IF THEN ELSE blocks, one inside another	☐	☐

G True or false

6 marks

Write T (true) or F (false) in the box.

1. ☐ Simple selection has one condition with two options.
2. ☐ In Flowchart 1, when the coin shows head, the computer shows "tails".
3. ☐ Selection out of many options selects one option out of more than two.
4. ☐ Scratch program 2 has six IF THEN ELSE blocks.
5. ☐ In Scratch program 2, the answer is kept in the variable Value.
6. ☐ When none of the five conditions is satisfied, the dice program shows 6.

CHAPTER 2

Repetition

? UNTIL THE TANK IS FULL

"Every morning Amma switches on our water pump."

"She switches it off only when the tank is full!"

Water flows into the tank **again and again** until a **condition** is satisfied: the tank is full. This is **repetition**.

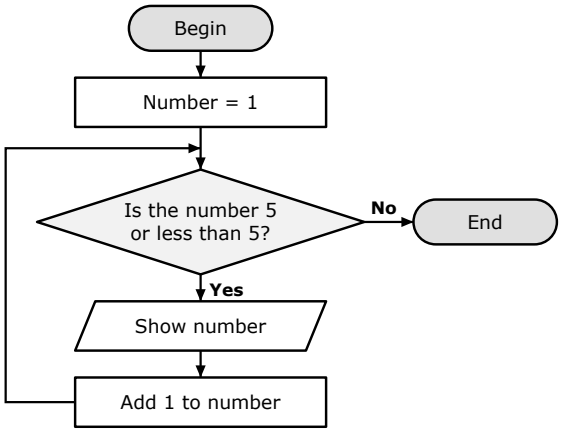


Colour me!

3.3 Control structure with repetition

Repetition is about an action getting **repeated again and again**. With repetition, both the **beginning** and the **end** are based on a **condition**.

Example 1: displaying the numbers 1 - 5



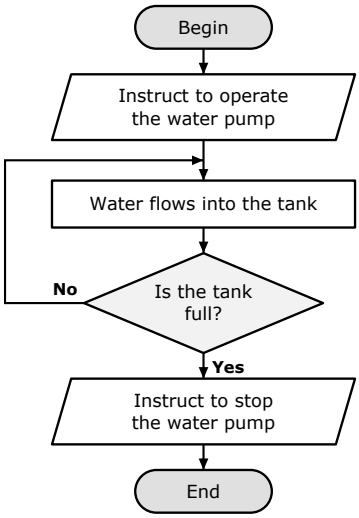
Flowchart 3: display numbers 1 - 5

i HOW FLOWCHART 3 WORKS

- The condition is: "Is the number 5 or less than 5?"
- At the beginning the condition is checked. It is true, so the repetitive work starts.
- The number is shown, 1 is added to it and the condition is checked again.
- When the number is more than 5, the condition is not satisfied and the repetition ends.

Example 2: filling a tank with water

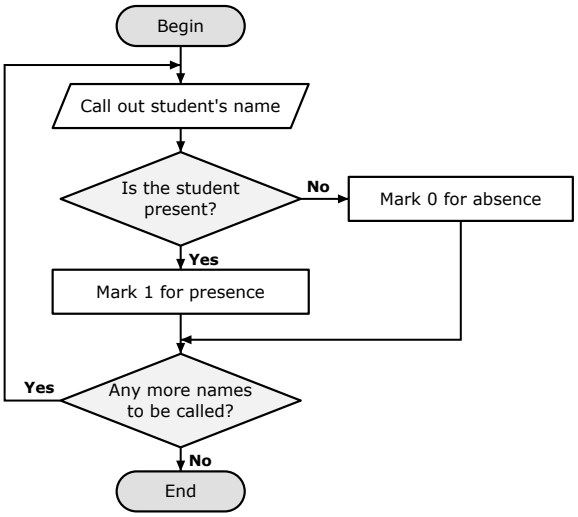
A water pump fills a tank. The pump is operated until the tank becomes full.



Flowchart 4: filling a tank with water

Example 3: marking the attendance register

If the student is present, the register is marked with 1. If the student is absent, it is marked with 0.

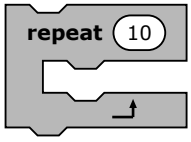
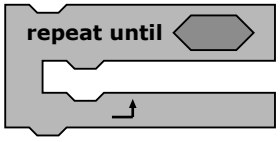
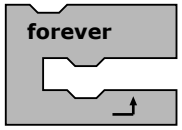
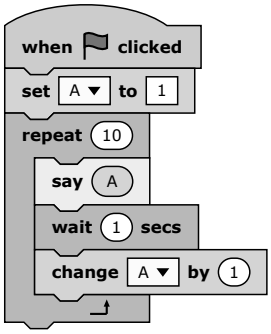
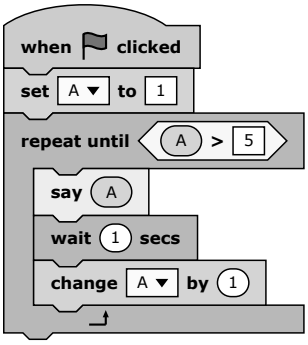
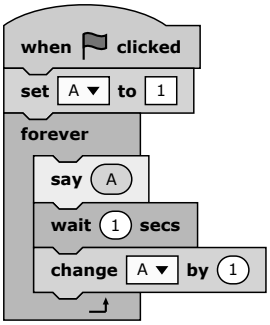


Flowchart 5: marking the attendance register

3.4 Scratch repetition control structures

Three repetition control structures are available in Scratch (Table 3.1). They are in **Control**. The blocks to repeat go inside the "mouth" of the block.

Table 3.1: repetition control structures

Repetition with a definite number of times	Repeating till a given condition is satisfied	Repeating an infinite number of times
		
The slot is for the number of times.	*The slot is for the condition.*	*No slot: it never stops.*
		
Scratch program 3: displaying the numbers 1 - 10 in 10 seconds	*Scratch program 4: displaying all positive integers 1 - 5*	*Scratch program 5: displaying all positive integers from 1*

DO IT - PRACTICAL 2 (WORKBOOK ACTIVITY 3.2)

Try the three repetition blocks (Scratch programs 3, 4 and 5). **Ask your teacher first.**

1	Open Scratch. Click **Data** and make a variable called **A**.
2	Build Scratch program 3: **set A to 1**, then **repeat 10** (Control) with **say A**, **wait 1 secs** and **change A by 1** inside it.
3	Click the green flag. Count the numbers that the cat says. How long does it take?
4	Take out **repeat 10**. Put **repeat until A > 5** in its place (Scratch program 4). Run it.
5	Now use **forever** (Scratch program 5). Run it. Click the red stop button to stop it.
6	Save each project with a clear name. Write what you saw in the table.

Program	The cat says ...	It stops ...
Scratch program 3		
Scratch program 4		
Scratch program 5		

"Try again and again until you get it right. That is repetition, and that is success."

- THE KING OF ICT

A Follow the flowchart

8 marks

Follow Flowchart 3 (in 3.3). The first row is done for you.

Number	5 or less than 5?	What is shown?
1	Yes	1
2		
3		
4		
5		
6		

1. How many numbers are shown?

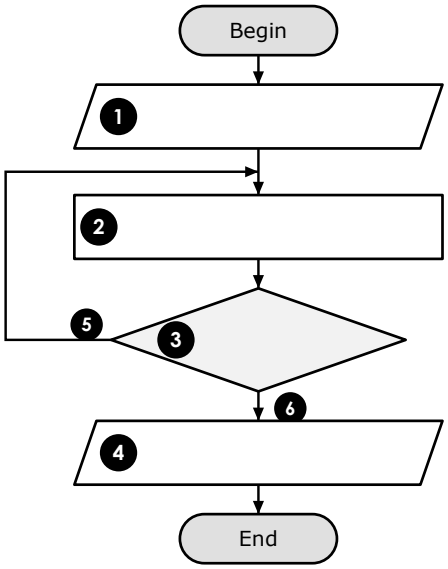
2. How many times is the condition checked?

3. What is the value of Number when the repetition ends?

B Complete the flowchart

6 marks

Flowchart 4: write the words in the numbered symbols and on the arrows 5 and 6.



USE THESE

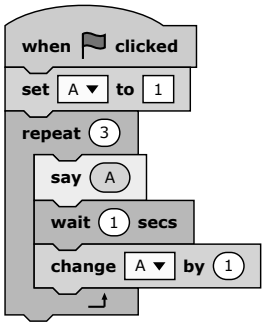
- Instruct to stop the water pump
- Is the tank full?
- Yes
- Instruct to operate the water pump
- No
- Water flows into the tank

C What will the cat say?

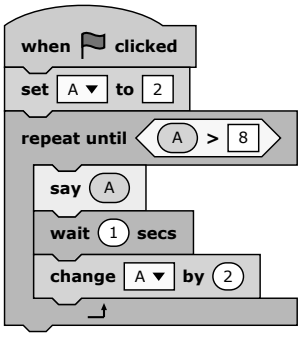
6 marks

Write the numbers that the cat says (2 marks each).

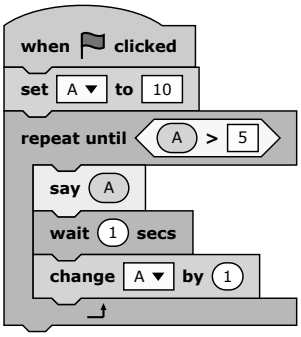
Script 1



Script 2



Script 3



D Mark the register

6 marks

Follow Flowchart 5 (in 3.3) for five students.

Name called out	Present or absent?	Mark
Kasun	present	
Dilini	absent	
Amal	present	
Nimal	absent	
Sachini	present	

1. How many times does the flowchart follow the **Yes** arrow of "Any more names to be called?"

E Which repeat block?

6 marks

Write the best block for each job: repeat, repeat until or forever.

You want to ...	Block
Say the numbers 1 to 10, one number each second.	
Say the numbers from 1 until A is greater than 5.	
Say all the positive integers from 1, without stopping.	
Ask for the names of five countries.	
Show a random number from 1 to 100 again and again, without end.	
Show the multiples of a number until Count is greater than 12.	

F True or false

6 marks

Write T (true) or F (false) in the box.

1. ☐ Repetition is about an action getting repeated again and again.

2. ☐ A repetition does not need a condition to begin or to end.

3. ☐ The repeat block repeats its instructions a definite number of times.

4. ☐ The forever block stops when A is greater than 5.

5. ☐ In Flowchart 4, the water pump stops when the tank is full.

6. ☐ In Flowchart 3, 1 is added to the number before the number is shown.

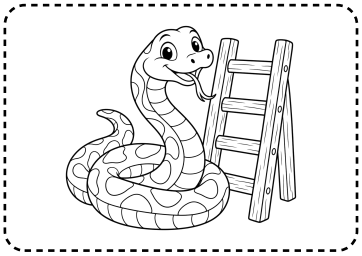
CHAPTER 3

Programs with Repetition

? A GAME THAT GOES ON AND ON

"Can the computer play Snakes and Ladders with us?"
"Yes! Turn after turn, until a counter reaches square 36."

In this chapter you build programs that repeat: multiples, random numbers, colourful patterns and a whole game.

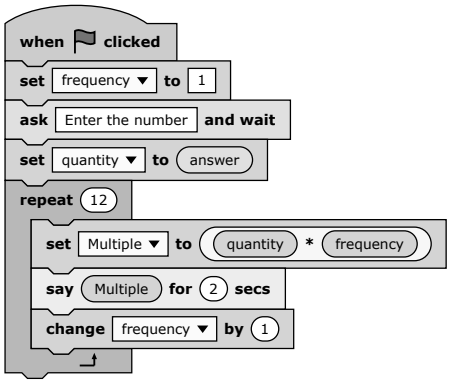


Colour me!

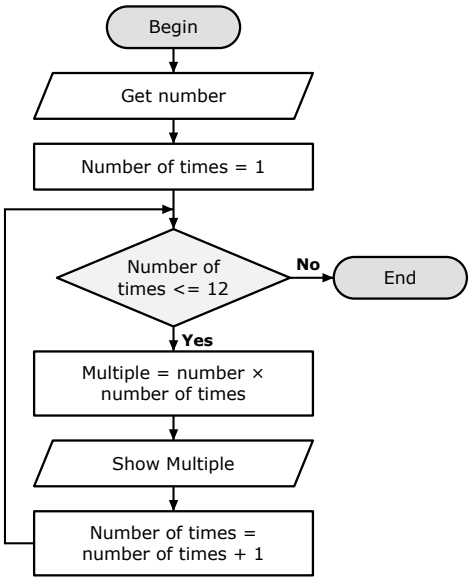
3.5 Developing visual programs involving repetition

Example 1: the first 12 multiples of any number

For 2 they are 2, 4, 6 ... 24; for 3 they are 3, 6, 9 ... 36. **Number of times** starts at 1 and grows by 1. At 13, **Number of times <= 12** (smaller than or equal to 12) is not satisfied and the repetition ends. Scratch program 6 uses **repeat 12**.



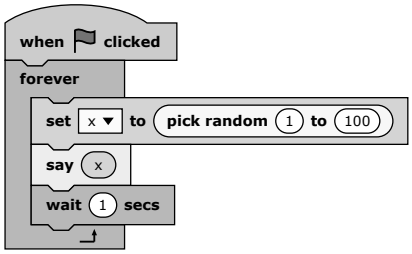
Scratch program 6



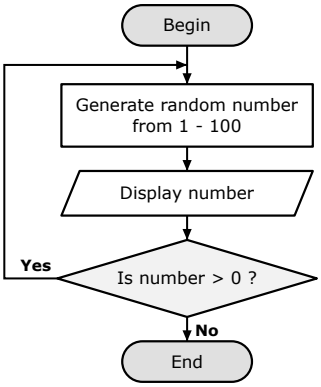
Flowchart 6: the first 12 multiples

Example 2: a random number from 1 to 100, again and again

The condition of Flowchart 7 always remains **true**; it never changes to false. Therefore **continuous repetition** takes place. Scratch program 7 uses **forever**.



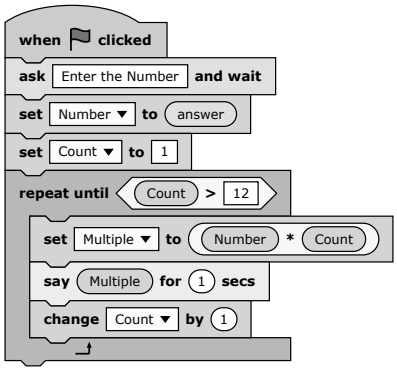
Scratch program 7



Flowchart 7: random numbers from 1 to 100

Example 3: the multiples, with repeat until

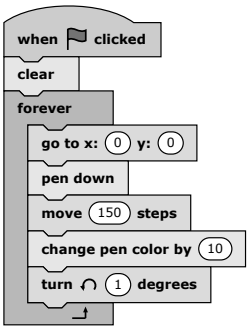
The same output as Flowchart 6. It repeats until **Count > 12**.



Scratch program 8

Example 4: a multi-coloured circle

Coloured lines from the centre, turning 1 degree each time, forever.



Scratch program 9

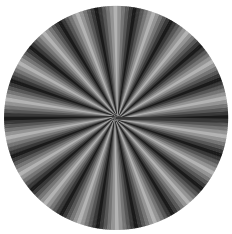


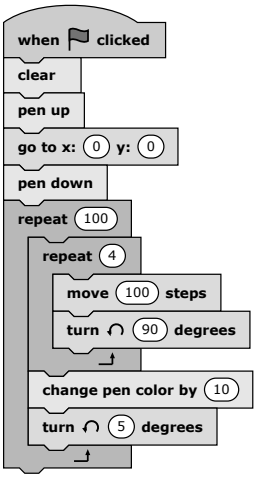
Figure 3.2

3.6 Programming with nested repetition

Here, there are **repetitions within repetitions**. A repetition within a repetition is called a **nested repetition**.

Example 1: a repetition within a repetition

The inner **repeat 4** draws one square: move 100 steps and turn 90 degrees, four times. The outer **repeat 100** draws 100 squares. After each square it changes the pen colour and turns 5 degrees.



Scratch program 10

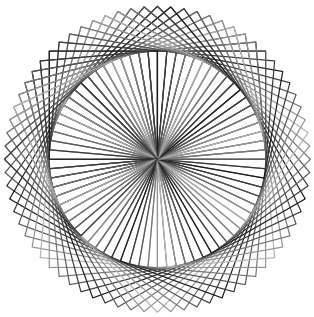


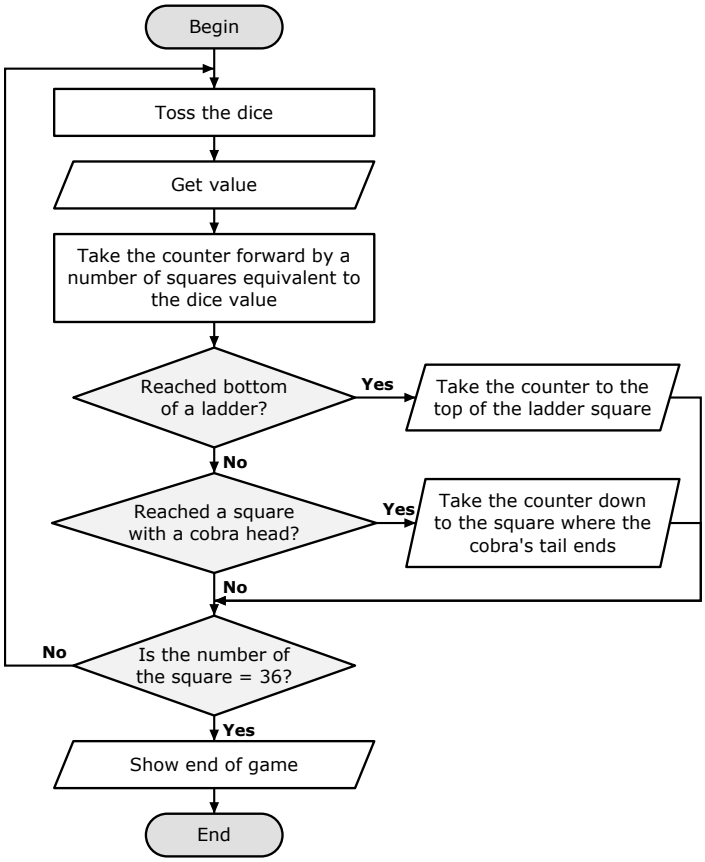
Figure 3.3

Example 2: repetitions with selection

Flowchart 8 is a complete Snakes and Ladders game. The game is **repeated**, turn after turn, until the number of the square is 36.

In every turn, two **selections** check: did the counter reach the bottom of a ladder? Did it reach a square with a cobra head?

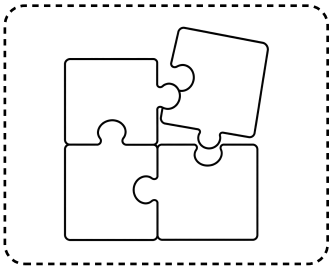
So the game has **repetitions with selections**. Its Scratch program (Scratch program 11) is in Practical 3.



Flowchart 8: the Snakes and Ladders game

"Small loops inside big loops make beautiful patterns.
Small efforts every day make champions."

- THE KING OF ICT

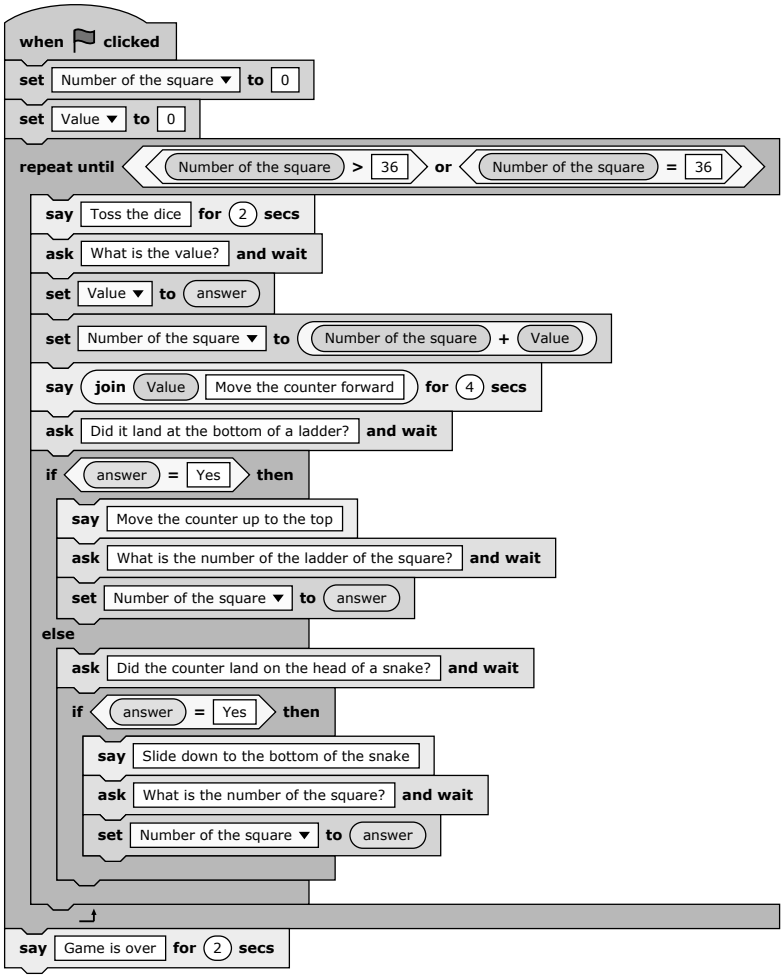


Colour me!

DO IT - PRACTICAL 3 (WORKBOOK ACTIVITY 3.3)

Nested repetition and the Snakes and Ladders game. **Ask your teacher first.**

1	Build Scratch program 10 (the **Pen** blocks are in **Pen**). Watch it draw the squares of Figure 3.3.
2	Now the game: make the variables **Number of the square** and **Value**. Build Scratch program 11 below. The **or** block is in **Operators**.
3	Play Snakes and Ladders with a friend on a real board. Click the green flag at the start of the game.
4	In every turn answer the cat: the value of the dice, **Yes** or **No**, and the number of the square.
5	Does the game end at square 36? Save the project as **Snakes and Ladders**.



Scratch program 11: the Snakes and Ladders game

"Follow the program step by step, like the computer does. Then you will never be lost."

- THE KING OF ICT



Colour me!

A Follow the program

8 marks

Scratch program 8: you type 7. The first row is done for you.

Count	Count > 12?	The cat says
1	No	7
2		
3		
4		
...	...	...
12		
13		

1. How many numbers does the cat say?

2. What is the value of Count when the repetition ends?

3. Scratch program 6 gives the same output. Which repetition block does it use?

B Count the repetitions

6 marks

Look at Scratch program 10 and answer.

Scratch program 10 (in 3.6)	Answer
How many times does the outer repeat block run?	
How many times does the inner repeat block run, each time?	
How many times does the **move 100 steps** block run altogether?	
How many degrees does the sprite turn to draw one square?	
How many squares does the program draw?	
Which block gives each square a new colour?	

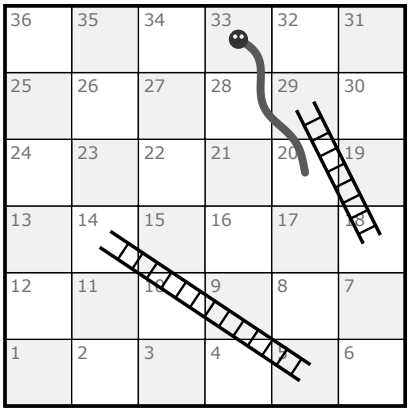
C Play the game

8 marks

Write the number of the square at the end of each turn.

Kamal plays with Scratch program 11. On his board the ladder at square 5 goes up to 14 and the ladder at square 18 goes up to 29. The snake at square 33 goes down to 20. Start: the number of the square is 0.

Turn	Dice value	Number of the square
1	3	
2	2	
3	4	
4	4	
5	6	
6	5	
7	6	



1. What does the cat say after the last turn?

"Every programmer makes mistakes. Good programmers find them and fix them."

- THE KING OF ICT

D True or false

6 marks

Write T (true) or F (false) in the box.

1.		A repetition within a repetition is called a nested repetition.
2.		In Flowchart 7, the condition "Is number > 0?" changes to false after some time.
3.		Scratch program 6 and Scratch program 8 give the same output.
4.		In Scratch program 10, the repeat 4 block is inside the repeat 100 block.
5.		Scratch program 9 stops after it draws 360 lines.
6.		The Snakes and Ladders program repeats until the number of the square is 36 or more.

E Answer these

4 marks

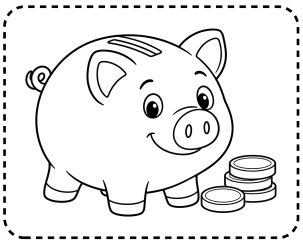
Write full sentences.

1. Why does the repetition in Flowchart 7 never end? (2 marks)

2. Scratch program 11 has repetition and selection. Write its repetition block and one selection block. (2 marks)

"You do not learn programming by reading. You learn it by doing."

- THE KING OF ICT



Colour me!

CHAPTER 4

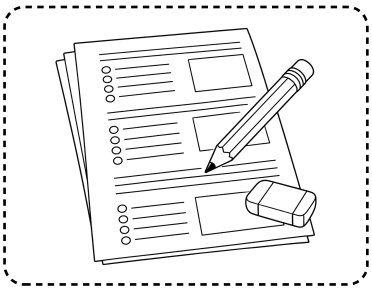
Arrays and Problem Analysis

? FIVE PAPERS, FIVE MARKS

"I got 78, 85, 69, 93 and 87 in the five general knowledge papers."

"Five variables for five marks? What about forty students?"

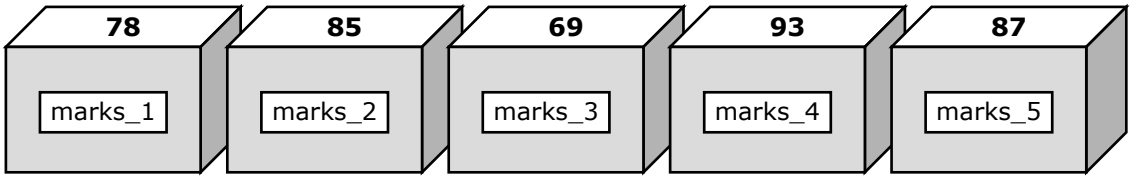
An **array** keeps many items with **one name**. In Scratch, an array is called a **list**.



Colour me!

3.7 Programming with arrays

To store values in a program we need **variables**: one variable for each value. For example, the marks of a student for five general knowledge papers need five variables, marks_1 to marks_5:



Each variable needs a name. With a large number of variables this is difficult, and the program becomes complicated and large. **Arrays** are used to get over this problem.

WHAT IS AN ARRAY?

An **array** is a **data structure** that can store any number of items using a **single variable name**. With arrays, programs become less complex and the number of instructions is reduced.

Building up arrays

Lists are used for arrays in Scratch. To build an array for the names of animals:

1. Select **Make a List** from **Data**.
2. Give the array a name: **Animal**.
3. Select **For this sprite only**.
4. Click **OK**.

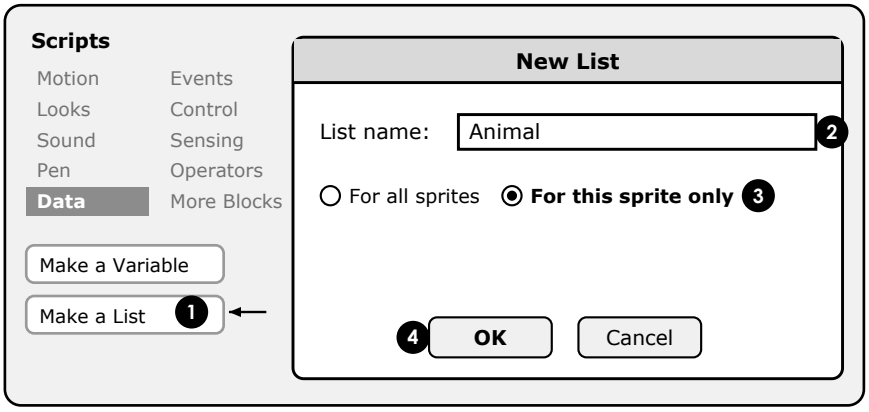


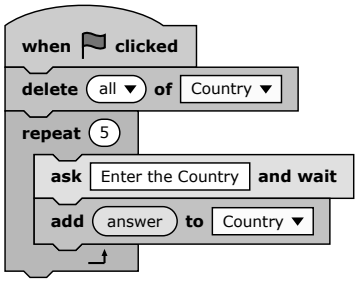
Figure 3.5: building up an array in Scratch

After these steps, the blocks for the **Animal** array appear in Data:

List block (Scratch program 12)	What it does
add thing to Animal	Adds an item at the end of the list.
delete 1 of Animal	Deletes an item (or all the items) of the list.
insert thing at 1 of Animal	Puts an item in at a place of the list.
replace item 1 of Animal with thing	Puts a new item in place of an old one.
item 1 of Animal	Gives the item at a place of the list.
length of Animal	Gives the number of items in the list.
Animal contains thing ?	Checks if an item is in the list: true or false.
show list Animal	Shows the list on the stage.
hide list Animal	Hides the list.

Assigning items into an array

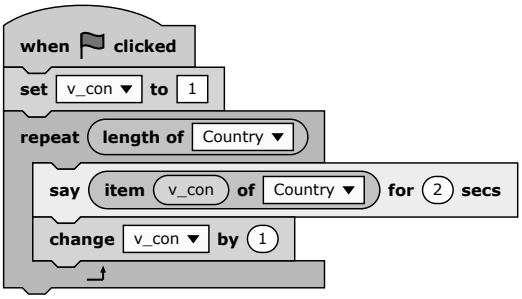
Scratch program 13 asks for five countries and adds each answer to the array **Country**.



Scratch program 13

Displaying the items of an array

Scratch program 14 says the items one by one. **v_con** is a variable; **Country** is the name of the array. The first item of a list is item 1.

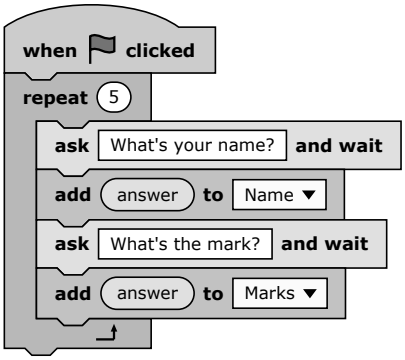


Scratch program 14

Variables or arrays?

Scratch program 15 keeps the names and marks of five students in ten variables (Name_1 ... Name_5, Marks_1 ... Marks_5). It needs 20 blocks under the hat block.

Scratch program 16 does the same job with **two arrays**, Name and Marks. With arrays we can use a **repetition**: this reduces the size of the program.



Scratch program 16: a program with repetition

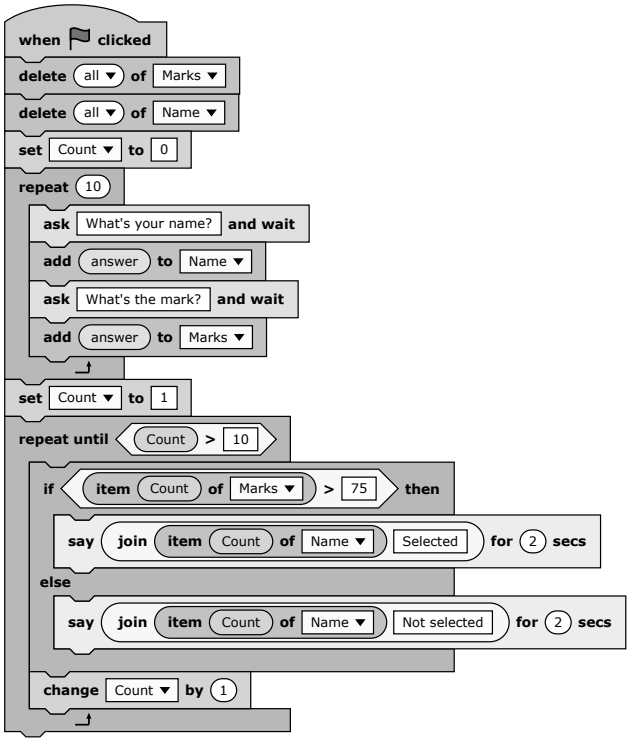
Sprite2: Name	
1	Nayana
2	Pawani
3	Radha
4	Paushi
5	Saman
+	length: 5

Sprite2: Marks	
1	74
2	65
3	87
4	42
5	60
+	length: 5

Selecting students

Ten students sat a general knowledge competition. Scratch program 17 keeps their names and marks in the arrays **Name** and **Marks**.

Then it checks every mark. If a mark is **greater than 75**, the cat says the name with "Selected". If not, it says the name with "Not selected".



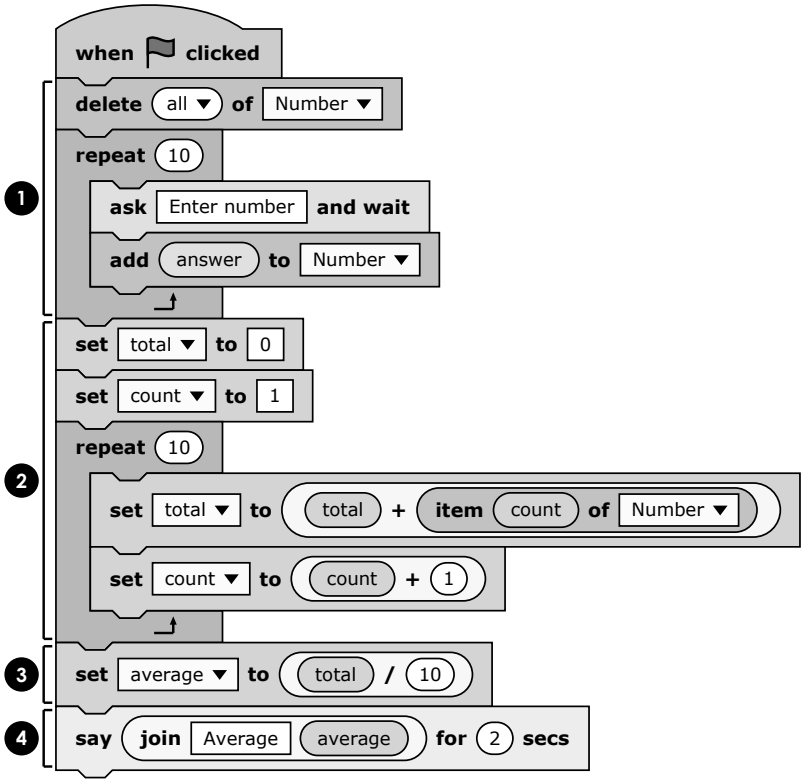
Scratch program 17

3.8 Programming with problem analysis

Dividing a problem into sections makes it easier to solve. Example: find the **average** of ten numbers. The problem has four sections:

1. Input ten numbers. *(Scratch program 18)*
2. Find the total of the ten numbers. *(Scratch program 19)*
3. Divide the total by ten to find the average. *(Scratch program 20)*
4. Output the average. *(Scratch program 21)*

Then the four sections are joined into one program: Scratch program 22. The numbers on its left show the four sections.



Scratch program 22: the average of ten numbers

DO IT - PRACTICAL 4 (WORKBOOK ACTIVITY 3.4)

Arrays in Scratch: countries and the average of ten numbers. **Ask your teacher first.**

1	Make a list called **Country**: Data, Make a List, For this sprite only, OK.
2	Build Scratch program 13. Run it and type the names of five countries. Watch the list on the stage.
3	Make a variable called **v_con**. Build Scratch program 14. Does the cat say all five countries?
4	Make the list **Number** and the variables **total**, **count** and **average**. Build Scratch program 22.
5	Run it with ten numbers. Check the average with a calculator. Save both projects.

My ten numbers	total	average

"One name, many items. Keep your data neat and your program short."

- THE KING OF ICT



Colour me!

A Read the lists

6 marks

Look at the two lists. Write the answers.

Sprite2: Name	
1	Nayana
2	Pawani
3	Radha
4	Paushi
5	Saman
+	length: 5

Sprite2: Marks	
1	74
2	65
3	87
4	42
5	60
+	length: 5

1. **item 3 of Name** gives .....
2. **item 5 of Marks** gives .....
3. **length of Marks** gives .....
4. **Name contains Saman?** gives .....
5. **Name contains Kamal?** gives .....
6. The mark of Pawani is .....

B What does each block do?

6 marks

Write the correct letter in the box.

A - The list block		B - What it does
1. **add** thing **to** Animal		**A.** Gives the number of items in the list.
2. **delete** 1 **of** Animal		**B.** Puts thing in at place 1. The other items move down.
3. **insert** thing **at** 1 **of** Animal		**C.** Adds thing at the end of the list.
4. **item** 1 **of** Animal		**D.** Checks if thing is in the list: true or false.
5. **length of** Animal		**E.** Deletes item 1 of the list.
6. Animal **contains** thing **?**		**F.** Gives item 1 of the list.

C Who is selected?

6 marks

Use the if block of Scratch program 17: item Count of Marks > 75.

Name	Mark	What does the cat say?
Nayana	74	
Pawani	65	
Radha	87	
Paushi	42	
Saman	60	

1. Kumari got exactly 75 marks. Is she selected? Why?

.....

.....

"Divide a big problem into small sections. Then solve it, one section at a time."

- THE KING OF ICT

D Follow the average program

6 marks

You run Scratch program 22 with these ten numbers. What is ...

Number	item 1	item 2	item 3	item 4	item 5	item 6	item 7	item 8	item 9	item 10
	4	7	5	8	6	9	3	7	5	6

1. the value of total after the first repetition?

2. the value of total after the third repetition?

3. the value of total at the end?

4. the value of count at the end?

5. the value of average?

6. the number of times the ask block runs?

E Sections of a problem

8 marks

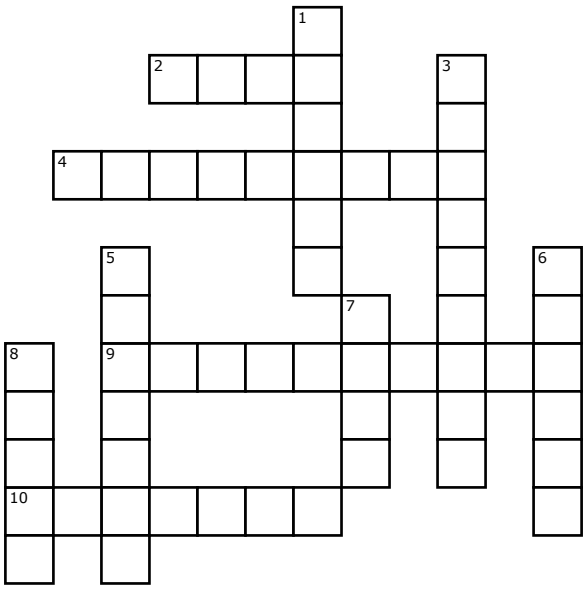
Number the sections in order. Write the Scratch program (18, 19, 20 or 21) of each.

Section of the average problem	Order (1 - 4)	Scratch program
Divide the total by ten to find the average.		
Output the average.		
Input ten numbers.		
Find the total of the ten numbers.		

F Crossword

10 marks

All the words are in this unit. Write one letter in each square.



ACROSS

2. Flowchart 2 shows the value of a tossed ... (4)
4. Choosing one option under a condition (9)
9. An action that is repeated again and again (10)
10. The total of ten numbers divided by ten (7)

DOWN

1. A repetition within a repetition is a ... repetition (6)
3. It decides when a repetition begins and ends (9)
5. This block repeats an infinite number of times (7)
6. Scratch program 7 shows a ... number from 1 to 100 (6)
7. In Scratch, an array is called a ... (4)
8. It stores many items using a single name (5)

"Save early, save often. A lost project is a lost hour."

- THE KING OF ICT



Colour me!

REVISION

Unit 3 in One Page

Read this the night before the test

Topic	The key facts
Simple selection	One condition, two options: head or tail. True: one option. False: the other option.
Out of many options	One option out of more than two. The dice: five conditions are checked one by one; if none is true, the value is 6.
IF THEN ELSE	One part runs when the condition is true, the else part when it is false. An IF THEN ELSE block can go inside the else part of another one.
Repetition	An action repeated again and again. A condition decides when it begins and when it ends.
Flowcharts	A repetition is drawn with an arrow that goes back up. The decision symbol checks the condition.
repeat	Repeats a definite number of times: repeat 10 says 1 - 10 in 10 seconds.
repeat until	Repeats till a given condition is satisfied: repeat until A > 5.
forever	Repeats an infinite number of times (its condition never becomes false).
Nested repetition	A repetition within a repetition: repeat 4 inside repeat 100 runs 400 times.
With selection	Snakes and Ladders: repeat until the square is 36 or more; IF blocks check the ladder and the snake.
Array	A data structure that stores any number of items with a single name. In Scratch it is a list.
List blocks	add, delete, insert, replace, item, length of, contains, show list, hide list.
Problem analysis	Divide the problem into sections: input, total, average, output. Then join them.

★ TICK WHEN YOU CAN DO IT

- [ ] I can explain simple selection and selection out of many options.
- [ ] I can read and draw flowcharts with repetition.
- [ ] I can use the repeat, repeat until and forever blocks.
- [ ] I can follow a program and write what the cat says.
- [ ] I can count how many times a nested repetition runs.
- [ ] I can make a list, add items to it and use its items.
- [ ] I can divide a problem into sections and build its program.

"Do your best work today. Tomorrow you will be proud of it."

- THE KING OF ICT

WORD POWER

Unit 3 Word List

Twenty-seven words you must know

Word	What it means
Selection	Choosing which steps to carry out by checking a condition.
Simple selection	Selecting one option out of two options under a condition.
Out of many options	Selecting one option out of more than two options.
Condition	A question whose answer is true or false (Yes or No).
Flowchart	A drawing of the steps of a program, with symbols joined by arrows.
Decision symbol	The diamond of a flowchart. It has a Yes exit and a No exit.
IF THEN ELSE	One part runs when the condition is true, the other part when it is false.
Control structure	Sequence, selection or repetition: it decides the flow of a program.
Repetition	A statement or a set of statements repeated again and again.
repeat	The Scratch block that repeats a definite number of times.
repeat until	The Scratch block that repeats till a given condition is satisfied.
forever	The Scratch block that repeats an infinite number of times.
Nested repetition	A repetition within a repetition.
Variable	A named place that keeps one value, like Count or total.
Multiple	A number times 1, 2, 3 ... The multiples of 3 are 3, 6, 9 ...
Random number	A number chosen by chance, like pick random 1 to 100.
Pen	Scratch blocks that draw a line as the sprite moves.
Sprite	A character on the Scratch stage, like the cat.
Data structure	A way to store data in a program, like an array.
Array	A data structure that stores any number of items with a single name.
List	An array in Scratch.
Make a List	The button in Data that builds a new list (an array) in Scratch.
Item	One value in a list: item 2 of Name is Pawani.
length of	The list block that gives the number of items in a list.
add	The list block that puts a new item at the end of a list.
Average	The total of the numbers divided by how many numbers there are.
Problem analysis	Dividing a problem into sections so that it is easier to solve.

"Learn the words today. Speak the language of programmers tomorrow."

- THE KING OF ICT

"A king is not born. A king is made, one practice at a time."

- THE KING OF ICT

END OF UNIT

Unit 3 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. Selecting "head" or "tail" when you toss a coin is a simple
   (a) repetition (b) selection (c) array (d) list
2. How many decision symbols does Flowchart 2 (the dice) have?
   (a) 1 (b) 2 (c) 5 (d) 6
3. An action that is repeated again and again is called
   (a) selection (b) repetition (c) an array (d) a sprite
4. In Flowchart 4, the water pump stops when the tank is
   (a) empty (b) full (c) open (d) small
5. In Flowchart 5, a student who is present is marked with
   (a) 0 (b) 1 (c) 5 (d) 12
6. For the number 3, the first 12 multiples end with
   (a) 12 (b) 24 (c) 33 (d) 36
7. Scratch program 9 (with forever) draws a multi-coloured
   (a) square (b) circle (c) list (d) cat
8. Dividing a problem into sections makes it
   (a) harder (b) easier (c) longer (d) slower

2 Fill in the blanks

8 marks

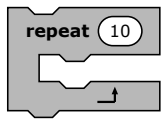
One word in each blank.

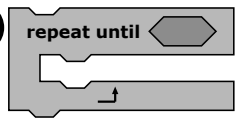
1. Simple selection selects one option out of ..... options.
2. In a flowchart, a condition is written in the ..... symbol.
3. With repetition, both the beginning and the end are based on a ..... .
4. Scratch program 3 says the numbers 1 to 10 in ..... seconds.
5. In Flowchart 7, the condition always remains ..... , so the repetition never ends.
6. An ..... is a data structure that stores many items using a single name.
7. In Scratch, an array is called a ..... .
8. Scratch program 22 finds the ..... of ten numbers.

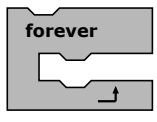
3 What does each block do?

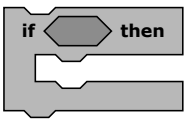
8 marks

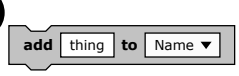
Write one short sentence under each block.

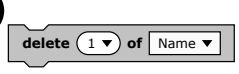
1. 

2. 

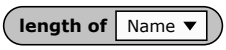
3. 

4. 

5. 

6. 

7. 

8. 

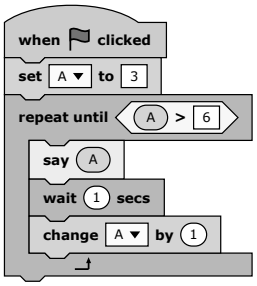
4 Answer these

16 marks

Write in the space given.

1. How is simple selection different from selection out of many options? (2 marks)

2. Look at the script. Write what the cat says. How many times does it repeat? (3 marks)



3. What is an array? Write two advantages of using arrays. (3 marks)

4. Draw a flowchart to display the numbers 1 to 10. (5 marks)

5. A repeat 4 block with a move block in it is inside a repeat 100 block. (a) What is this called? (b) How many times does the move block run? (3 marks)